

Total No. of Questions : 12]

SEAT No. :

P187

[5871] - 709

[Total No. of Pages : 2

B.E. (Computer Engg.)

**SOFTWARE TESTING AND QUALITY ASSURANCE
(Semester-I) (2015 Pattern) (410245B) (Elective-II)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Assume Suitable Data if necessary. Mention your assumptions.
- 2) Right indicates the full marks and Bifurcation for sub questions.
- 3) Draw suitable diagrams and tables if necessary.

Q1) What is Software Quality? Explain continual improvement cycle with neat labeled diagram of PDCA framework. **[6]**

OR

Q2) a) Explain customer's view of quality.

b) Define following terms:

- i) Defect
- ii) Benchmarking
- iii) Metrics

[6]

Q3) a) What do you mean by following software testing terms: Bug, Error and Mistake?

b) What is Test Plan Document? List and explain any two components of the Test Plan document.

[7]

OR

Q4) a) What are the skill set required by Software Tester?

b) Write short note on Mutation testing?

[7]

Q5) a) Explain Test automation framework with suitable diagram.

b) Differentiate between Automation testing and Manual testing.

[3+4=7]

OR

Q6) a) Explain features of good testing.

b) List and explain requirements for test tool.

[3+4=7]

P.T.O.

- Q7)** a) What is Selenium? What are different features of Selenium IDE?
b) List automation tools for software testing. Describe QTP in detail.
c) What are selenium test design considerations?

[6+5+5=16]

OR

- Q8)** a) State & explain the components of Selenium tool.
b) Write short note on Selenium Grid.
c) Differentiate between Selenium Web Driver and Selenium RC.

[6+5+5=16]

- Q9)** a) What is SQA? List various elements of SQA & describe any two in detail.
b) State & explain Principles of Quality management.
c) What is ISO standard? What are its advantages?

[6+6+5=17]

OR

- Q10)** a) What is Six Sigma? Explain the terms DMAIC & DMADV.
b) What are different quality dimensions suggested by Garvin?
c) How cost and risk factors are affecting software quality.

[6+6+5=17]

- Q11)** a) Write short note on Total Quality Management (TQM).
b) Explain following terms (any two)
i) Checklists
ii) Histogram
iii) Run Charts
c) Describe in detail Defect Injection & Defect Removal activities for a development process.

[6+6+5=17]

OR

- Q12)** a) Enumerate Ishikawa's seven basic quality tools. Explain any two in detail.
b) Describe key elements of Total Quality Management.
c) Explain with example Product Quality Metric.

[6+6+5=17]

